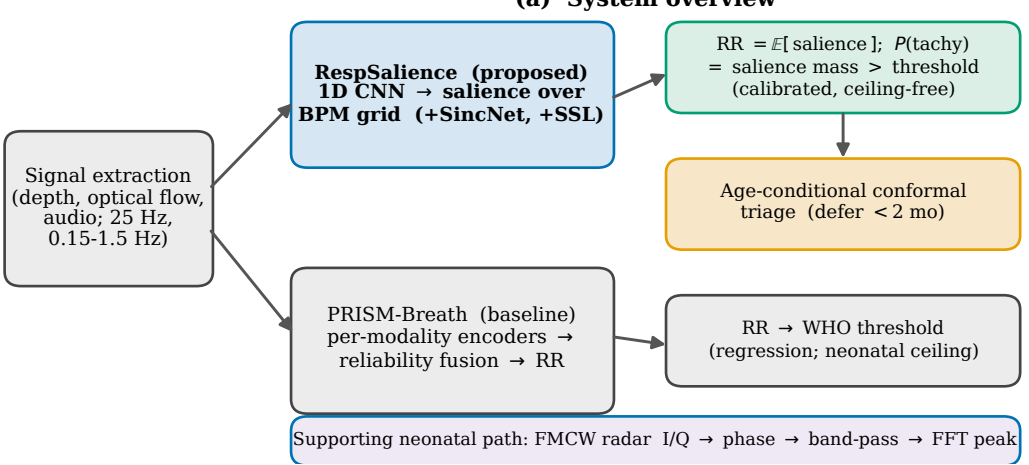
(a) System overview
Signal extraction (depth, optical flow, audio; 25 Hz, 0.15-1.5 Hz)
RespSalience (proposed) 1D CNN → salience over BPM grid (+SincNet, +SSL)
RR = 𝔼[salience]; P(tachy) = salience mass > threshold (calibrated, ceiling-free)
Age-conditional conformal triage (defer < 2 mo)
PRISM-Breath (baseline) per-modality encoders → reliability fusion → RR
RR → WHO threshold (regression; neonatal ceiling)
Supporting neonatal path: FMCW radar I/Q → phase → band-pass → FFT peak